University of BRISTOL

Object Detection

Viola-Jones Detector

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Object Detection Techniques

- ✓ **Line and circle detection**: Techniques like the Hough Transform can be used to detect lines and circles in an image, which can indirectly help locate objects with specific geometric shapes.
- ✓ **Colour-based detection**: In some cases, objects can be detected based on their colour properties. This is especially useful when objects have distinct and consistent colors.
- ✓ **Template matching**: Using sliding a template over the input image and finding regions where the template best matches the local image content.
- ✓ **Classifiers with sliding window detectors**: Applying image classification on overlapped patches in the image.
- ✗ **Deep learning-based object detectors**: Object detector automatically learns image features required for detection tasks, and instance segmentation.

(out of scope in this unit)

Classifiers with sliding window detectors

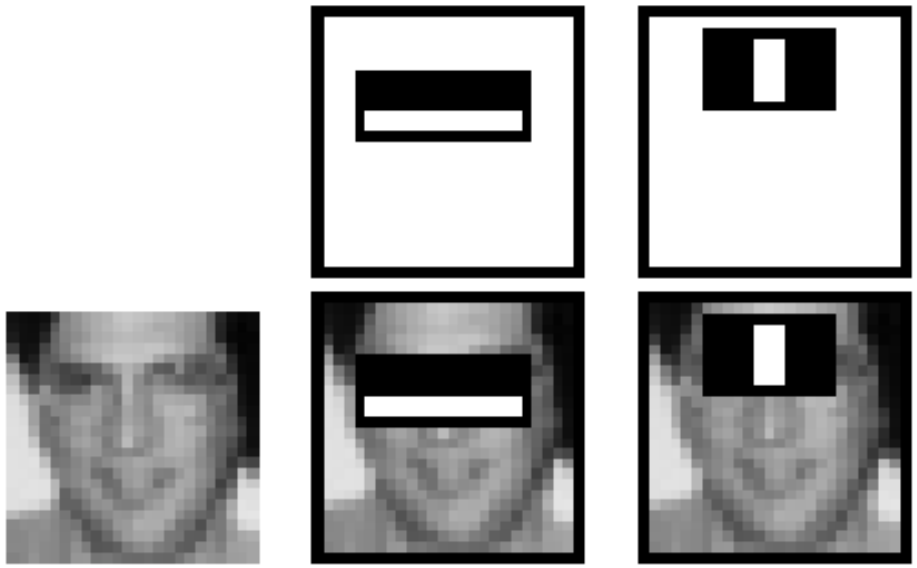
- Example Algorithm: **Viola & Jones' Real-time Method**
  - Sliding Window Detectors
  - Haar-like Features
  - Feature Extraction and Integral Images
  - Weak Classifiers
  - Boosting and Classifier Evaluation
  - Cascades of Boosted Classifiers



Best description of full details available in consolidated paper by
Viola and Jones, International Journal of Computer Vision, 2004

Haar-like Features

Viola & Jones' (2001)



filter 1

-1	-1	1	1
-1	-1	1	1
-1	-1	1	1
-1	-1	1	1

hard edge
feature1 = 2040

0	0	255	255
0	0	255	255
0	0	255	255
0	0	255	255

soft edge
feature1 = 1245

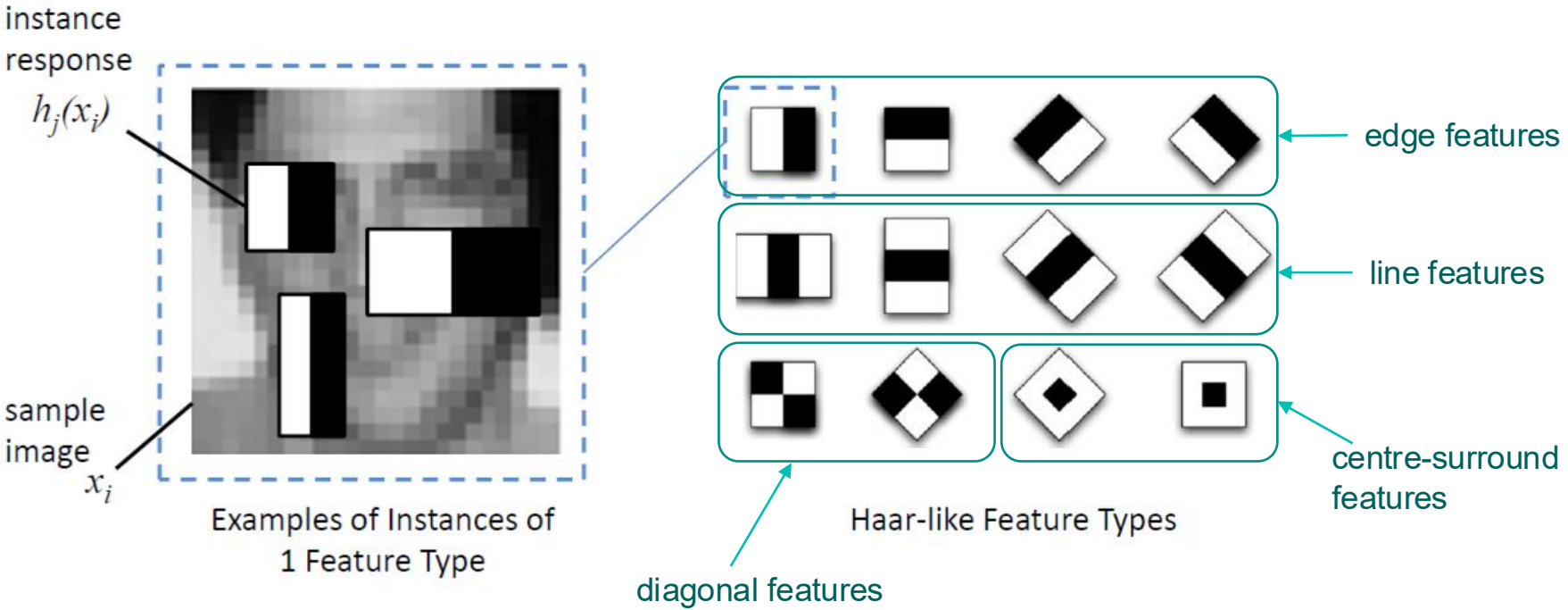
20	120	220	230
25	125	205	225
25	105	220	234
24	110	215	250

no edge
feature1 = 17

218	230	220	230
200	230	205	225
220	234	220	244
210	250	215	250

feature = sum of white pixels – sum of black pixels

Haar-like Features



Integral Images

I Image

	0	1	2	3	4	5	6	7
0	1	1	1	2	3	1	2	1
1	1	2	0	0	0	3	1	1
2	1	1	1	1	1	2	3	1
3	1	1	1	0	0	0	0	0
4	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	1	1

$\sum$

II Integral Image

	0	1	2	3	4	5	6	7
0	1	2	3	5	8	9	11	12
1	2	5	6	8	11	15	18	20
2	3	7	9	12	16	22	28	31
3	4	9	12	15	19	25	31	34
4	4	9	12	15	19	25	31	34
5	4	9	12	15	19	25	32	36

(IMAGE INTEGRATION)

$$\mathbf{II}(-1, y) = 0; \qquad \mathbf{II}(x, y) = \mathbf{II}(x-1, y) + A(x, y);$$

$$A(x, -1) = 0; \qquad A(x, y) = A(x, y-1) + \mathbf{I}(x, y).$$

x=0, y=0: A(0,0) = A(0,-1) + I(0,0) = 0 + 1 = 1
II(0,0) = II(-1,0) + A(0,0) = 0 + 1 = 1

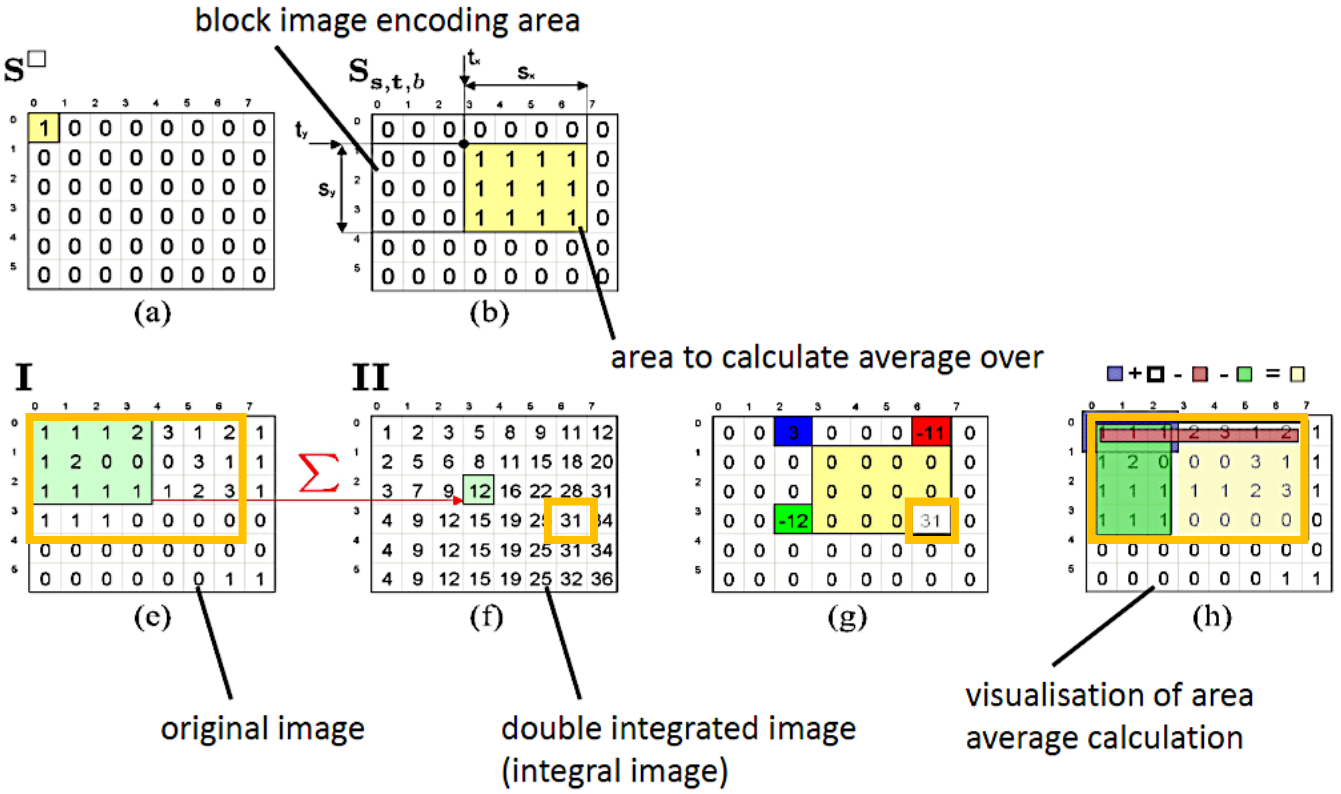
x=0, y=1: A(0,1) = A(0,0) + I(0,1) = 1 + 1 = 2
II(0,1) = II(-1,1) + A(0,1) = 0 + 2 = 2

x=1, y=0: A(1,0) = A(1,-1) + I(1,0) = 0 + 1 = 1
II(1,0) = II(0,0) + A(1,0) = 1 + 1 = 2

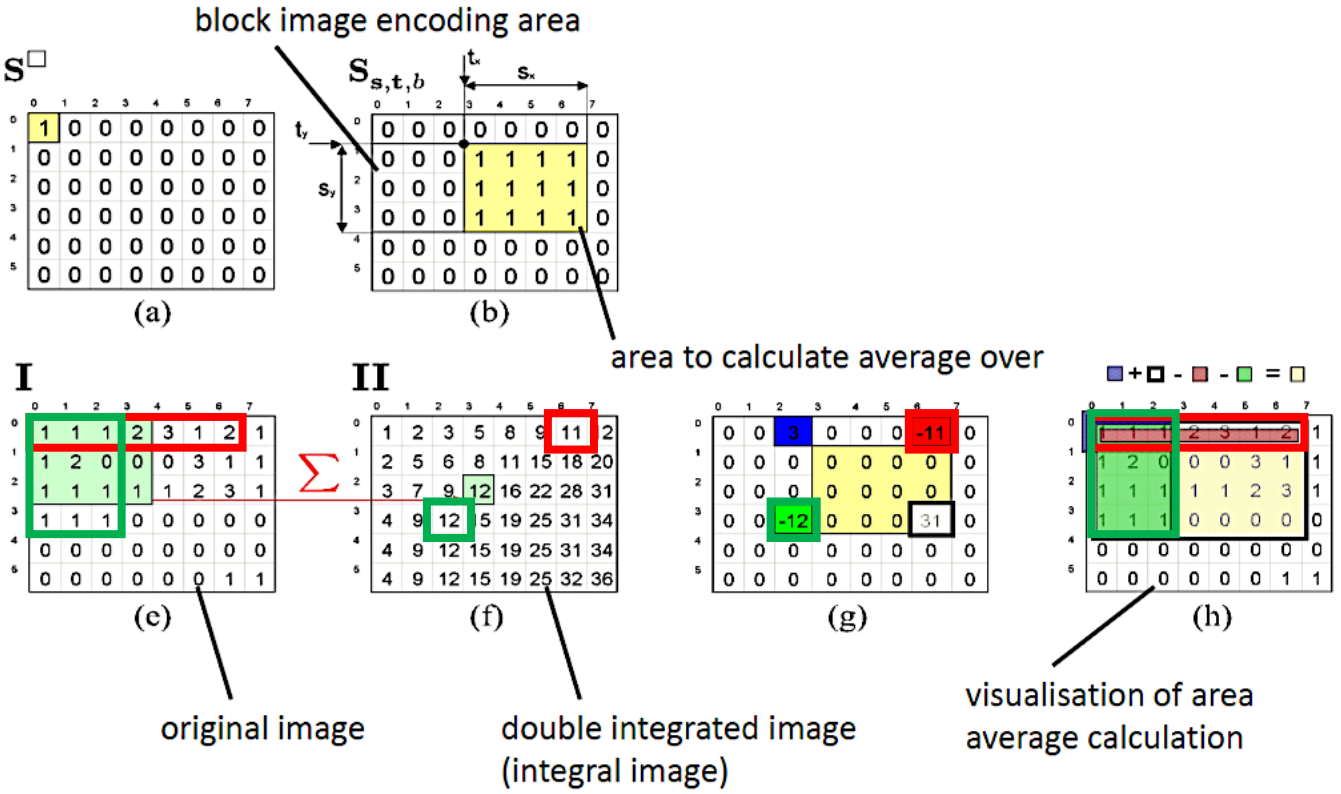
x=1, y=1: A(1,1) = A(1,0) + I(1,1) = 1 + 2 = 3
II(1,1) = II(0,1) + A(1,1) = 2 + 3 = 5

…

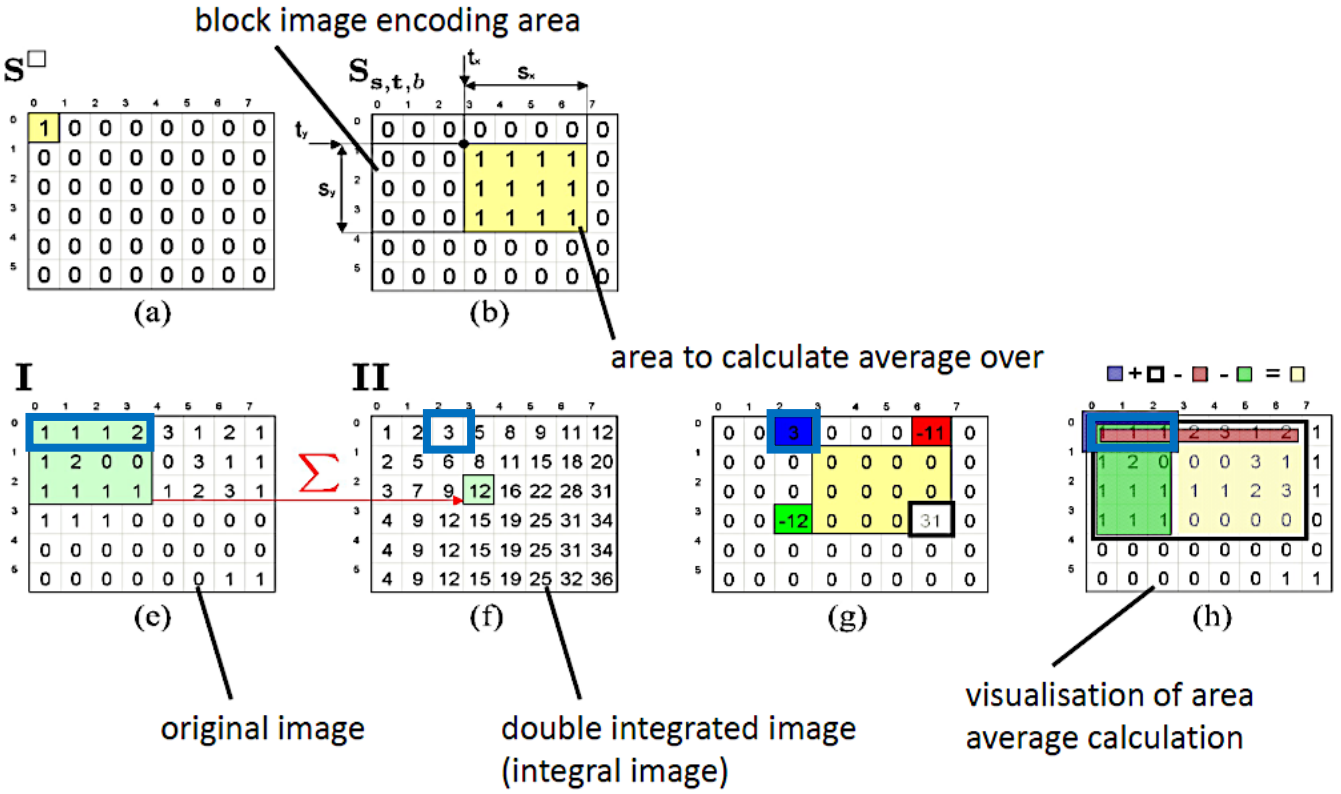
Calculating the Avg Pixel Value of Large Block Fast



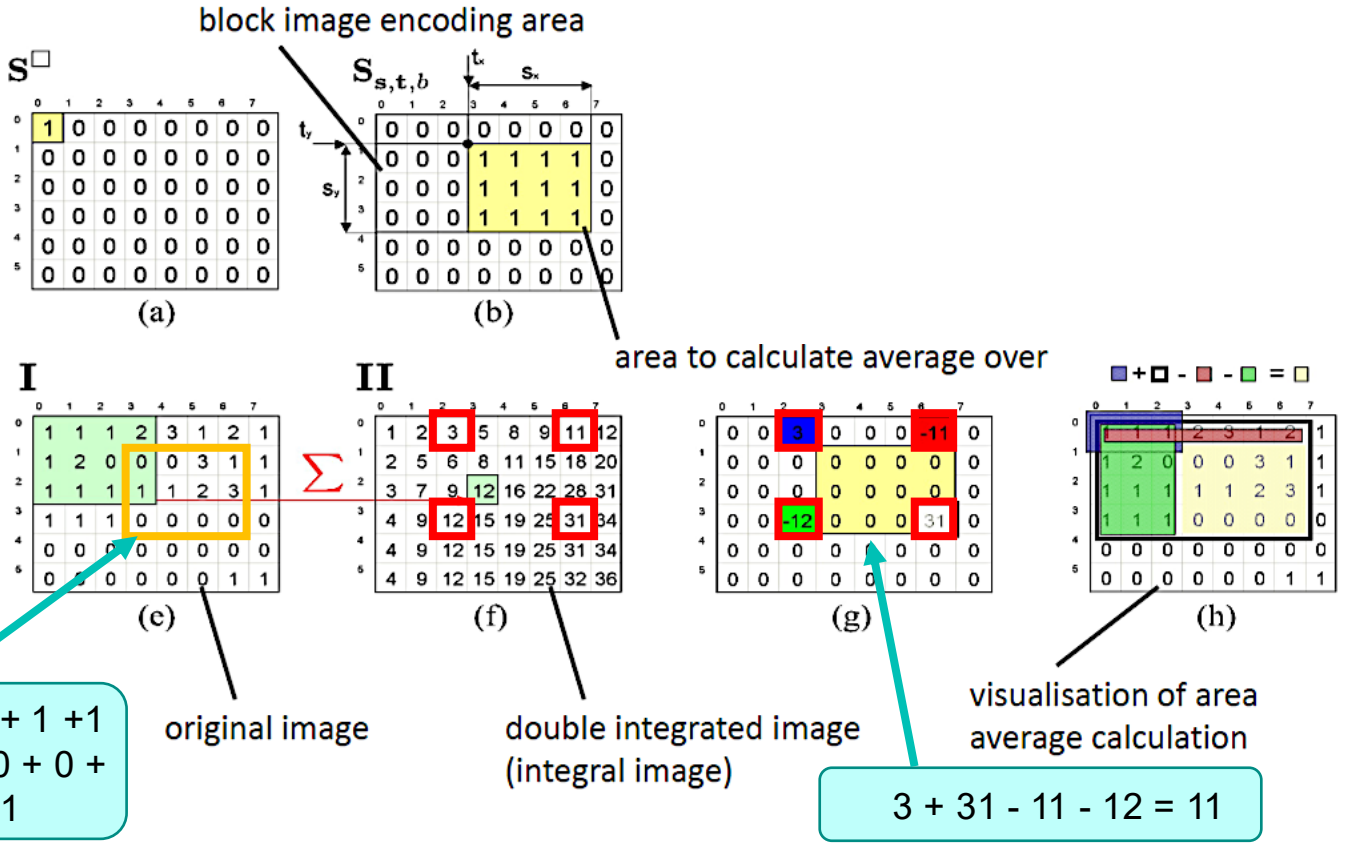
Calculating the Avg Pixel Value of Large Block Fast



Calculating the Avg Pixel Value of Large Block Fast



Calculating the Avg Pixel Value of Large Block Fast



Feature Extraction

x=2, y=2

y \ x	0	1	2	3	4	5	6	7	8	9
0	4	1	1	0	10	0	1	0	0	5
1	3	5	7	1	4	1	8	9	4	0
2	6	4	5	0	1	0	7	8	3	2
3	2	3	4	5	1	1	6	3	4	5
4	9	9	10	1	5	3	1	4	5	4
5	2	2	1	7	4	0	2	1	5	6
6	8	1	1	4	2	3	2	3	1	0
7	0	1	7	0	3	5	6	3	4	1
8	1	5	6	3	5	9	10	4	2	0
9	1	8	3	4	6	3	6	3	3	5

Image

y \ x	0	1	2	3	4	5	6	7	8	9
0	4	5	6	6	16	16	17	17	17	22
1	7	13	21	22	36	37	46	55	59	64
2	13	23	36	37	52	53	69	86	93	100
3	15	28	45	51	67	69	91	111	122	134
4	24	46	73	80	101	106	129	153	169	185
5	26	50	78	92	117	122	147	172	193	215
6	34	59	88	106	133	141	168	196	218	240
7	34	60	96	114	144	157	190	221	247	270
8	35	66	108	129	164	186	229	264	292	315
9	36	75	120	145	186	211	260	298	329	357

Integral Image

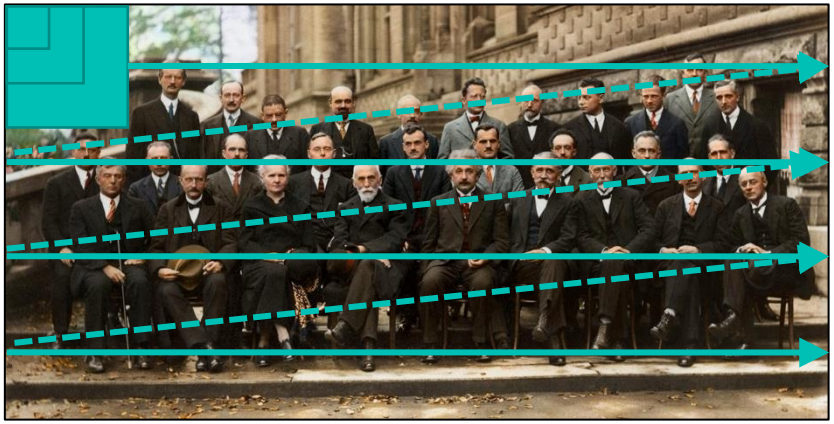
-1	-1	-1	1	1	1
-1	-1	-1	1	1	1
-1	-1	-1	1	1	1
-1	-1	-1	1	1	1
-1	-1	-1	1	1	1
-1	-1	-1	1	1	1

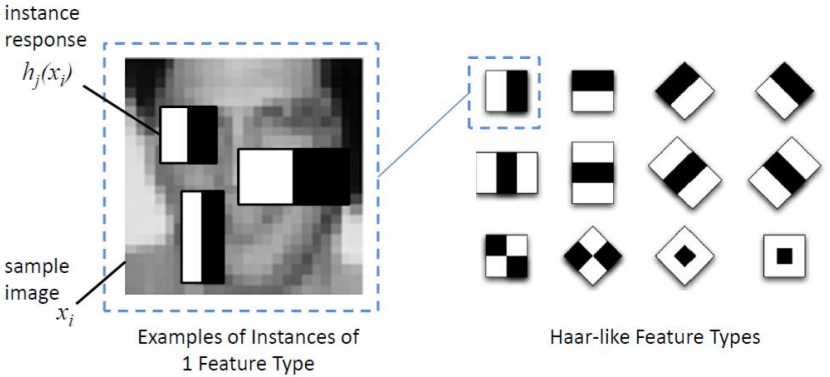
Haar filter

-5-0-1-4-5-1-10-1-5-1-7-4-1-4-2-7-0-
3+0+7+8+1+6+3+3+1+4+0+2+1+3+2+
3+5+6+3 = **-3**

- (144+13-36-60) + (221+36-55-144) = **-3**

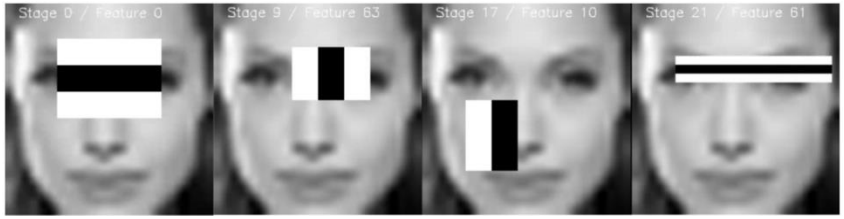
Viola & Jones' Real-time Method





Examples of Instances of 1 Feature Type

Haar-like Feature Types

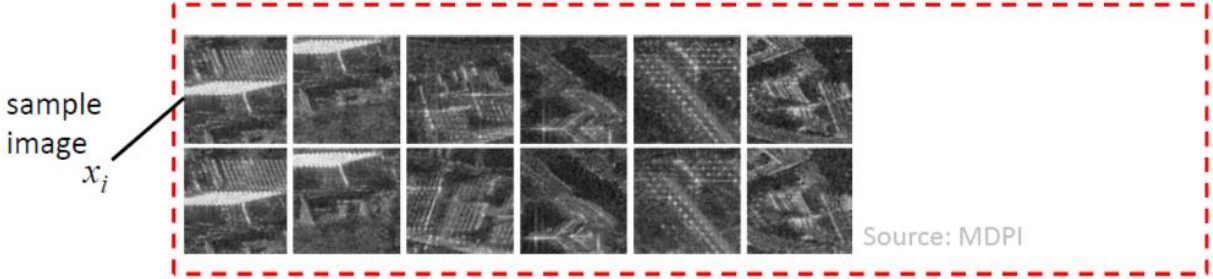


https://medium.com/@Andrew_D./computer-vision-viola-jones-object-detection-d2a609527b7c

Training

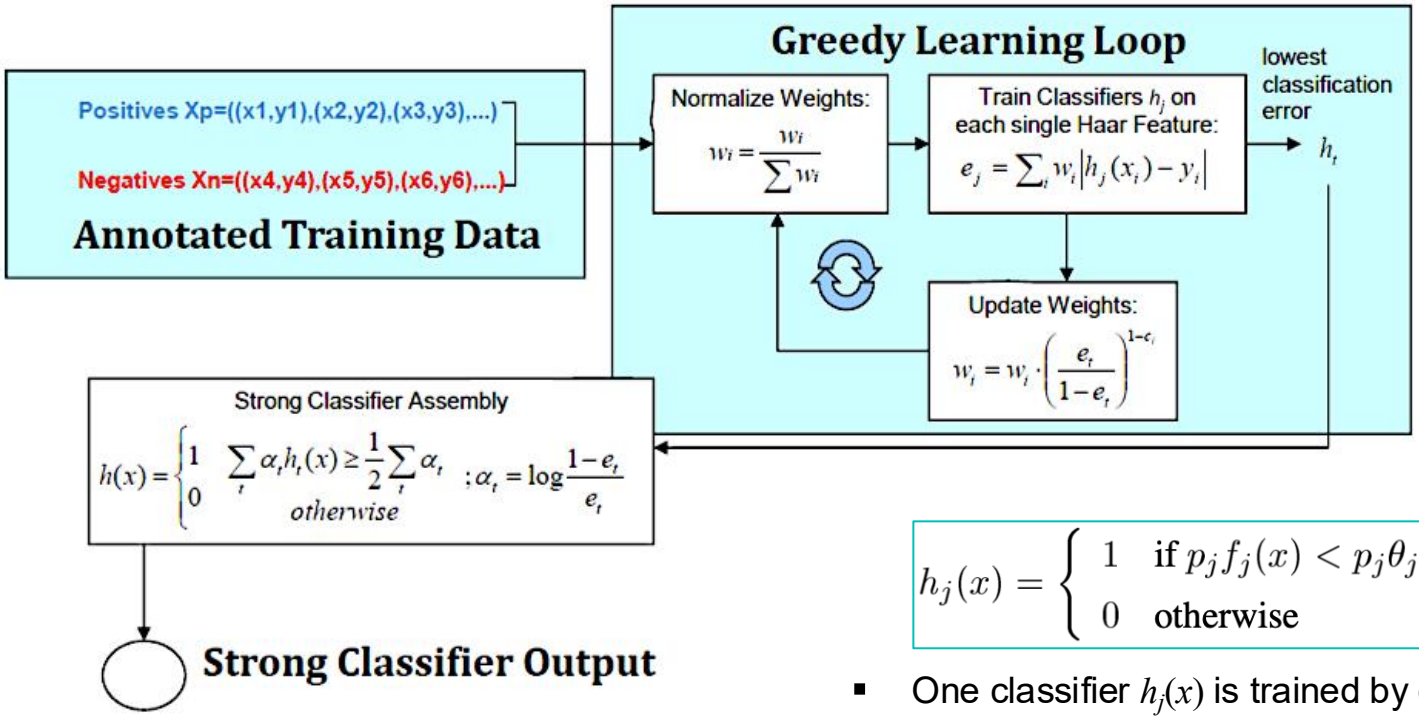


Positive Samples (e.g. FACE) … $(x_i, y_i = 1)$, $w_i = 1$



Negative Samples (e.g. NO-FACE) … $(x_i, y_i = 0)$, $w_i = 1$

AdaBoost Classifier

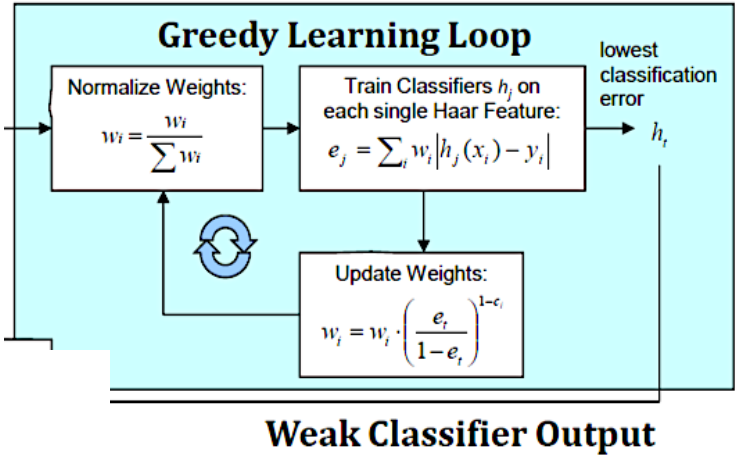


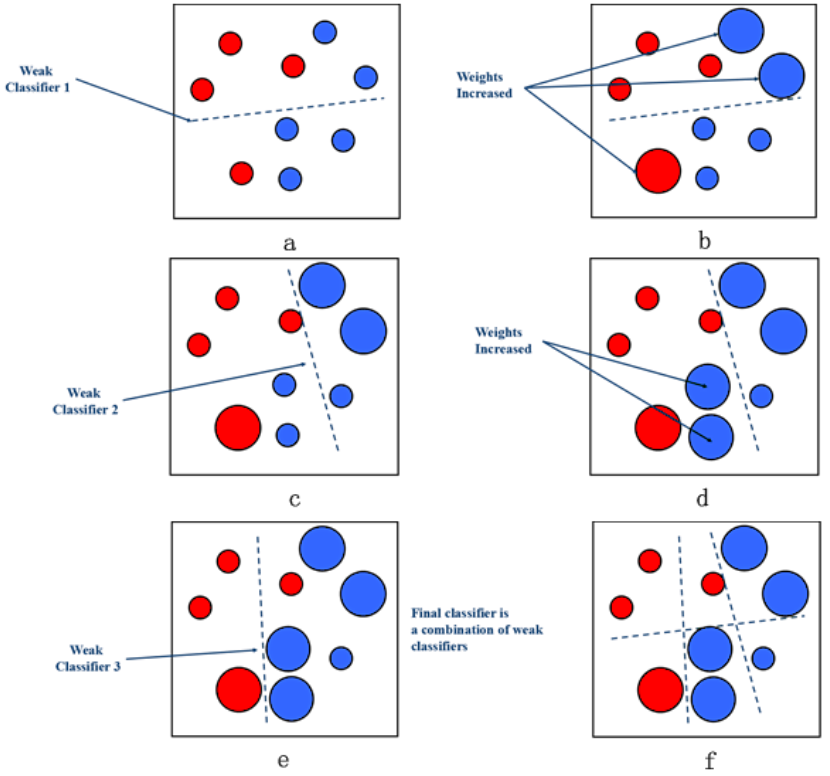
(also see paper by Viola and Jones 2004)

$$h_j(x) = \begin{cases} 1 & \text{if } p_j f_j(x) < p_j \theta_j \\ 0 & \text{otherwise} \end{cases}$$

- One classifier $h_j(x)$ is trained by one feature $f_j(x)$
  - θ_j is a threshold
  - p_j is a sign + or -

AdaBoost Classifier





https://www.codeproject.com/Articles/4114375/Step-by-Step-Guide-to-Implement-Machine-Learning

Adaboost Algorithm

- Given example images $(x_1, y_1), \ldots, (x_n, y_n)$ where $y_i = 0, 1$ for negative and positive examples respectively.
- Initialize weights $w_{1,i} = \frac{1}{2m}, \frac{1}{2l}$ for $y_i = 0, 1$ respectively, where m and l are the number of negatives and positives respectively.

- For $t = 1, \ldots, T$:
  1. Normalize the weights,
  $$w_{t,i} \leftarrow \frac{w_{t,i}}{\sum_{j=1}^{n} w_{t,j}}$$
  so that w_t is a probability distribution.
  2. For each feature, j, train a classifier h_j which is restricted to using a single feature. The error is evaluated with respect to w_t, $\epsilon_j = \sum_i w_i \left| h_j(x_i) - y_i \right|$.
  3. Choose the classifier, h_t, with the lowest error ϵ_t.
  4. Update the weights:
  $$w_{t+1,i} = w_{t,i} \beta_t^{1-e_i}$$
  where $e_i = 0$ if example x_i is classified correctly, $e_i = 1$ otherwise, and $\beta_t = \frac{\epsilon_t}{1-\epsilon_t}$.

- The final strong classifier is:

$$h(x) = \begin{cases} 1 & \sum_{t=1}^{T} \alpha_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^{T} \alpha_t \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha_t = \log \frac{1}{\beta_t}$

Non-Maximum Suppression (NMS)

- A post-processing step to remove redundant detections.



Non-Maximum Suppression (NMS)

- A post-processing step to remove redundant detections.

Algorithm 1 Non-Maximum Suppression Algorithm

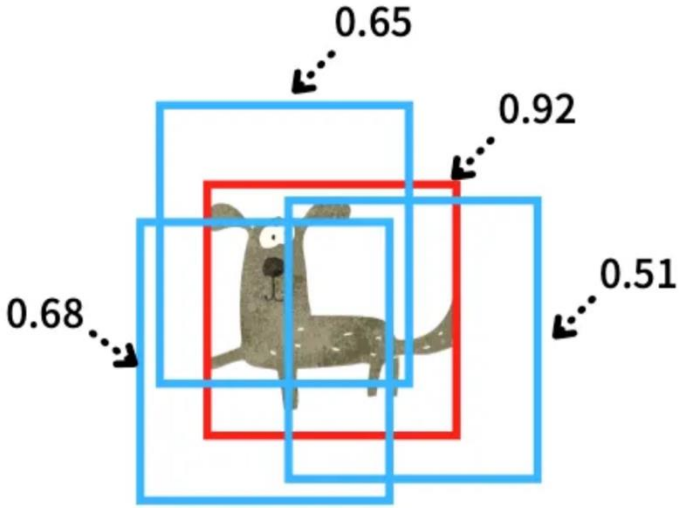
Require: Set of predicted bounding boxes B, confidence scores S, IoU threshold τ, confidence threshold T
Ensure: Set of filtered bounding boxes F

$F \leftarrow \emptyset$
Filter the boxes: $B \leftarrow \{b \in B \mid S(b) \geq T\}$
Sort the boxes B by their confidence scores in descending order
while $B \neq \emptyset$ **do**
    Select the box b with the highest confidence score
    Add b to the set of final boxes F: $F \leftarrow F \cup \{b\}$
    Remove b from the set of boxes B: $B \leftarrow B - \{b\}$
    for all remaining boxes r in B **do**
        Calculate the IoU between b and r: $iou \leftarrow IoU(b, r)$
        if $iou \geq \tau$ **then**
            Remove r from the set of boxes B: $B \leftarrow B - \{r\}$
        end if
    end for
end while

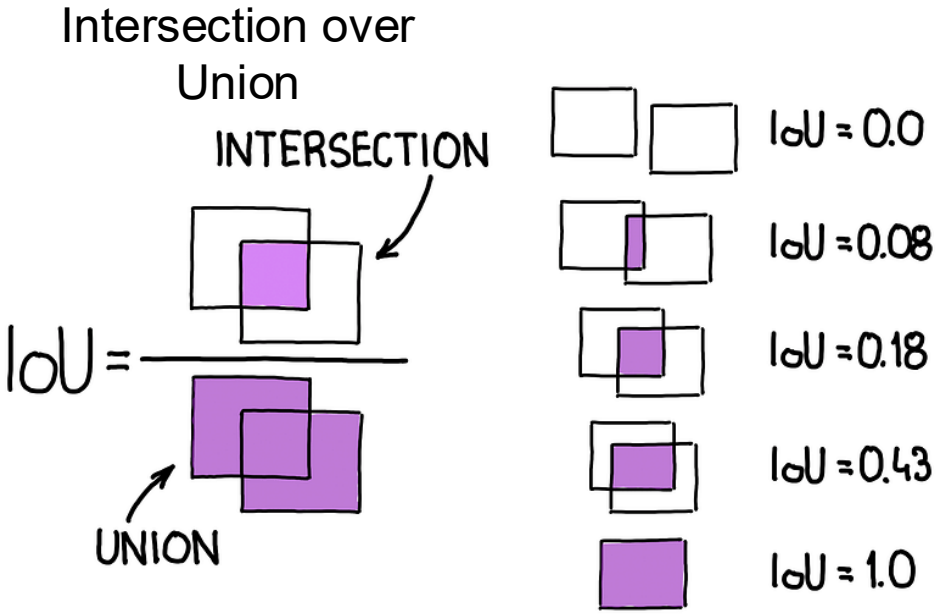
https://browse.arxiv.org/pdf/2304.00501.pdf



Non-Maximum Suppression (NMS)



1. take the largest probability box



2. Remove others with IoU score > threshold value

Non-Maximum Suppression (NMS)

1. Filter Bounding Boxes by Probability:

2. Sort Bounding Boxes by Probability:

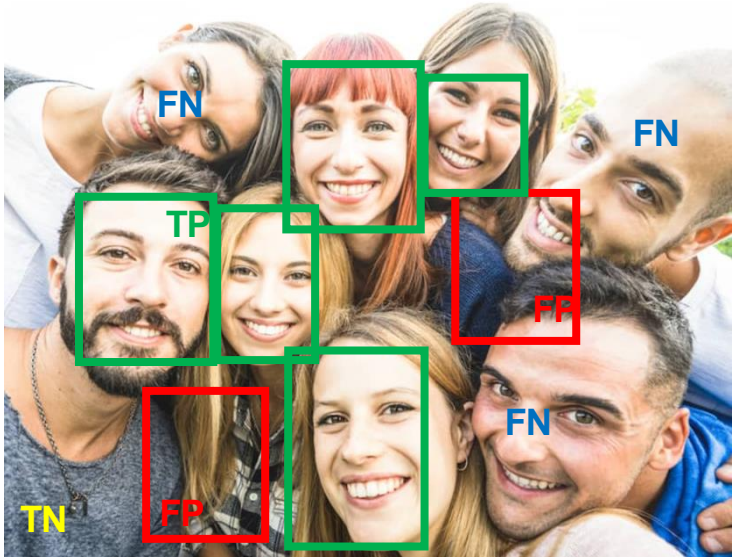
3. Non-Maximum Suppression Loop:



Performance Considerations

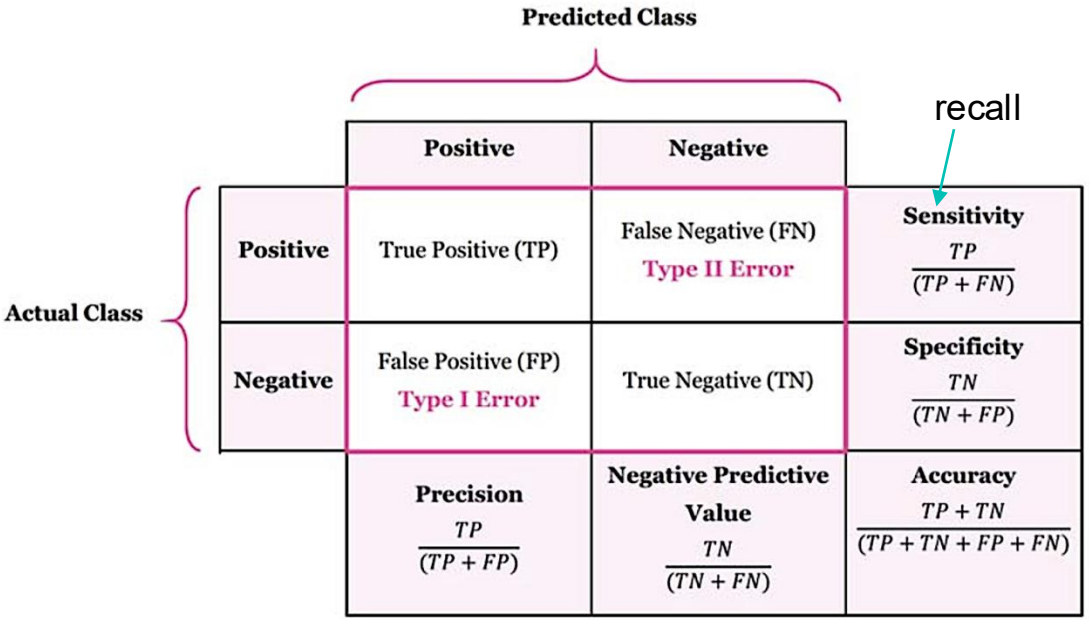
Actual Class \ Predicted Class	Positive	Negative	
Positive	True Positive (TP)	False Negative (FN) Type II Error	**Sensitivity** $\frac{TP}{(TP+FN)}$
Negative	False Positive (FP) Type I Error	True Negative (TN)	**Specificity** $\frac{TN}{(TN+FP)}$
	Precision $\frac{TP}{(TP+FP)}$	**Negative Predictive Value** $\frac{TN}{(TN+FN)}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

https://medium.com/@m.virk1/classification-metrics-65b79bfdd776



https://www.quickanddirtytips.com/articles/people-or-persons/

Performance Considerations

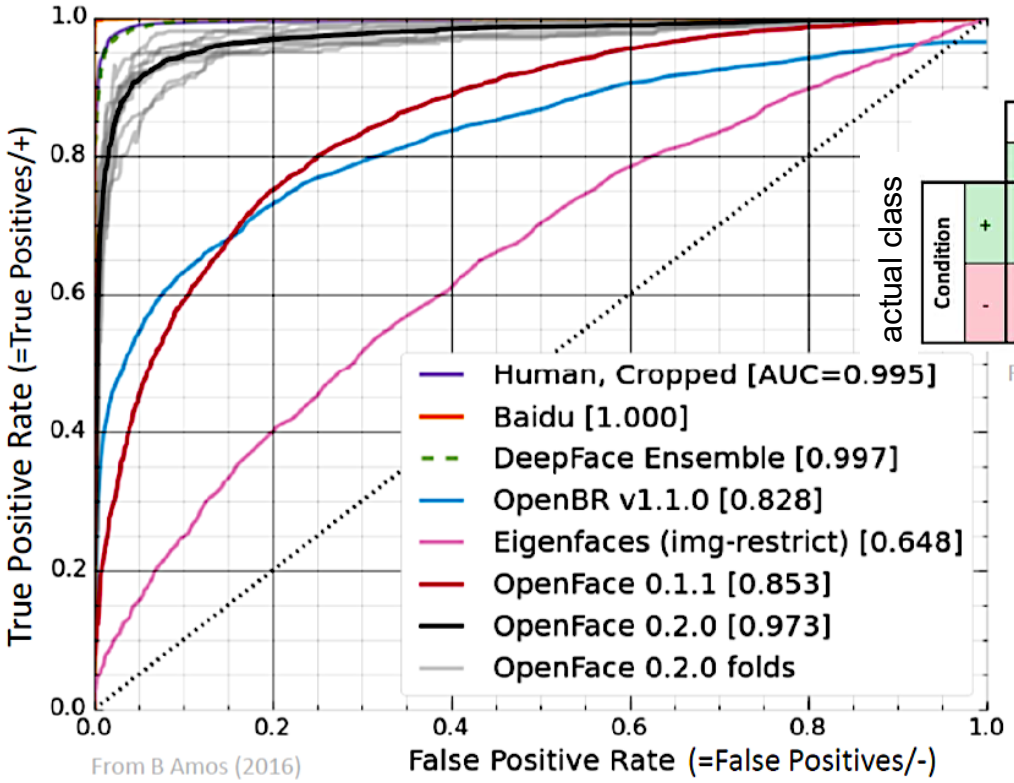


https://medium.com/@m.virk1/classification-metrics-65b79bfdd776

F-score

$$F_1 = 2\frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{2\text{tp}}{2\text{tp} + \text{fp} + \text{fn}}$$

Receiver Operating Characteristic (ROC) Curve

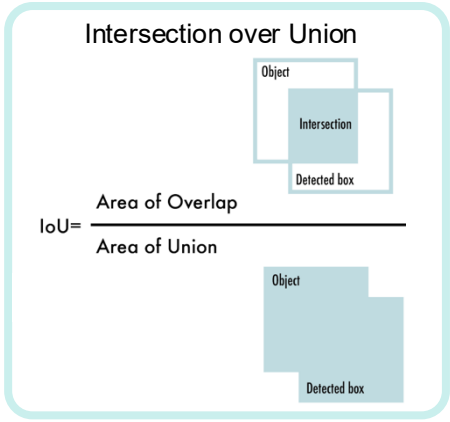


From B Amos (2016)

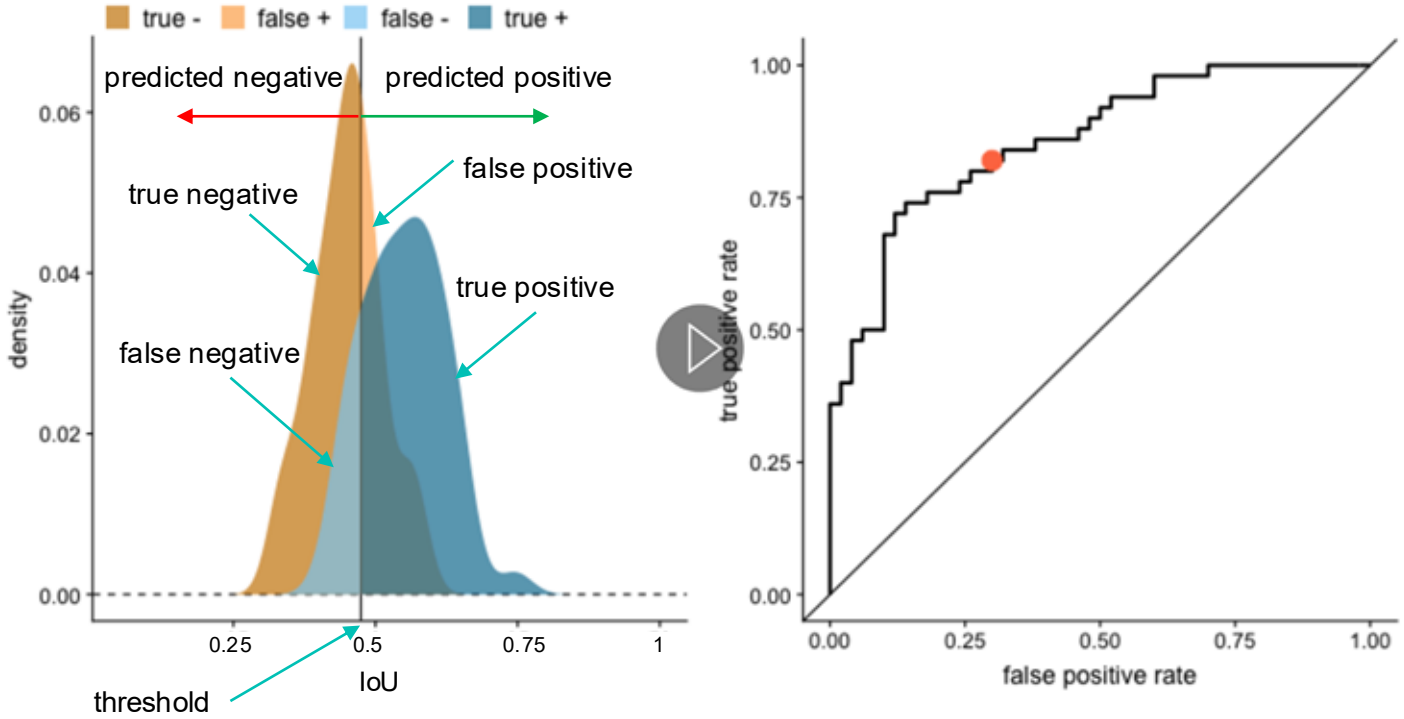
predicted class

actual class		Classification: Positive	Classification: Negative
Condition	+	True Positive	False Negative
	-	False Positive	True Negative

From R Wingate (2016)



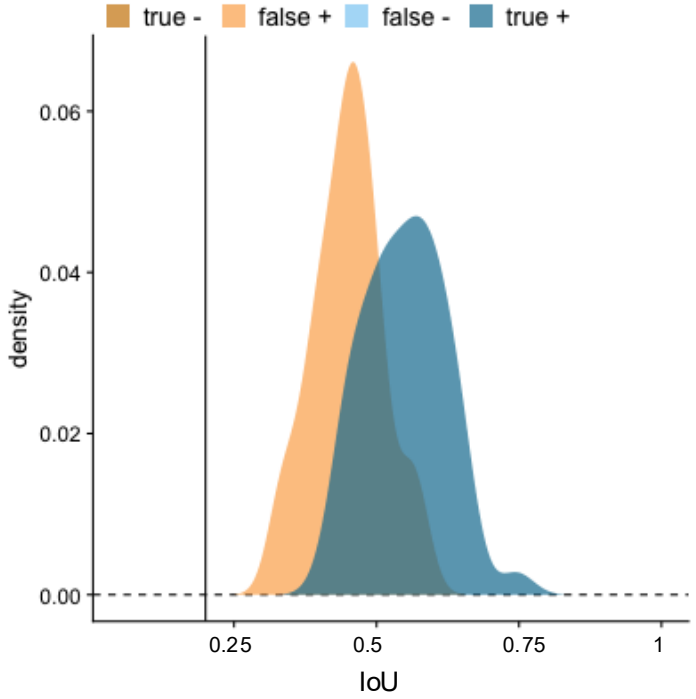
Receiver Operating Characteristic (ROC) Curve

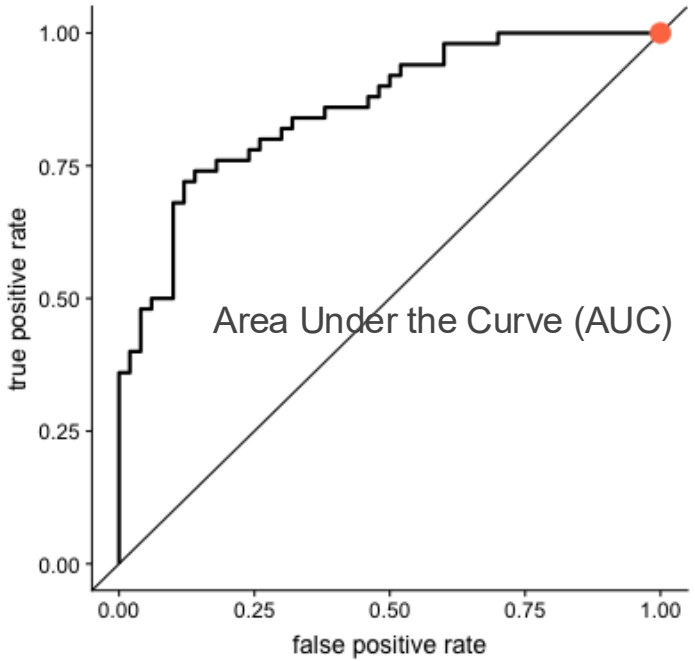


blue: actual positives (e.g. faces)
yellow: actual negatives (e.g. background)

https://paulvanderlaken.com/2019/08/16/roc-auc-precision-and-recall-visually-explained/

Receiver Operating Characteristic (ROC) Curve

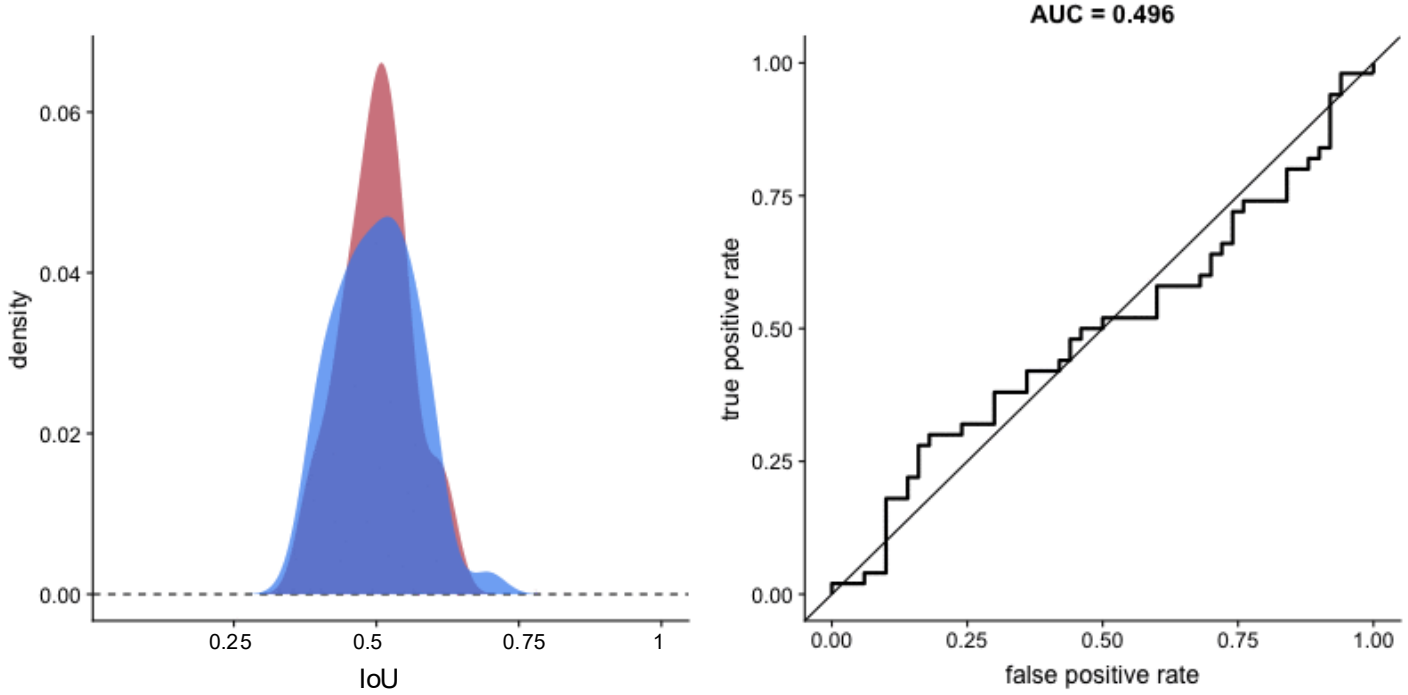




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https://paulvanderlaken.com/2019/08/16/roc-auc-precision-and-recall-visually-explained/

Receiver Operating Characteristic (ROC) Curve



https://paulvanderlaken.com/2019/08/16/roc-auc-precision-and-recall-visually-explained/

Viola–Jones Today

Still in use

- OpenCV Haar cascades (cv2.CascadeClassifier) ship pre-trained face and eye detectors
- Runs in real time on a CPU: good for low-power and embedded devices
- No GPU, no large training set needed at run time

Superseded for accuracy

- CNN detectors (e.g. MTCNN, RetinaFace, YOLO family) cope far better with pose, occlusion and lighting
- Features are learned from data rather than hand-designed Haar boxes
- Deep learning detectors are out of scope for this unit

Ideas that carried over: sliding windows → anchor boxes, cascades → early rejection / coarse-to-fine, plus NMS, IoU and precision–recall evaluation, all still used by modern detectors.

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Next

Motion

